



Unit 4 · Outcome 1 · Climate Change

Where does the Sun's energy go?

KK02

VCE Environmental Science · Units 3 & 4 · Exam study summary

Digital Vector EnviroSci · single-topic mastery summary with worked good-vs-bad answers, exam traps, command words and practice.



Where does the Sun's energy go?

KK02 Unit 4 · Climate Change · Dot point 2 Where does the Sun's energy go ? Every second, Cape Wirreanda Climate Observatory is bathed in sunlight. Some of that energy bounces straight back to space off the snow on the ranges. Some is soaked up by the dark Southern Ocean. Some is radiated back out as heat — only to be caught by gases in the air and sent back down. This dot point is the story of those three fates: reflected , absorbed , and re-emitted . 3D · LIVE Cape Wirreanda — watch sunlight split three ways 3D view unavailable — the concepts below wor

Earth runs on a solar budget

01 The big picture Earth runs on a solar budget Think of incoming sunlight like income . Earth can't store sunlight forever — over the long run, the energy it takes in must balance the energy it sends back out to space. That balance is what sets the planet's temperature. Mess with either side and the temperature shifts. The Sun sends energy as shortwave radiation — mostly visible light, high-energy, short wavelength. When sunlight reaches Earth, it meets one of three fates: ~30% Reflected Bounced straight back to space by clouds, the atmosphere, and brig

The albedo effect

02 Reflected energy The albedo effect Albedo is the proportion of incoming shortwave (solar) radiation that a surface reflects , on a scale from 0 to 1 (or 0% to 100%). A perfect black absorber has an albedo of 0; a perfect white mirror has an albedo of 1. High-albedo surfaces stay cool because they bounce energy away; low-albedo surfaces warm up because they soak it in. Surface Typical albedo Reflectivity Fresh snow 0.80–0.90 Sea ice 0.50–0.70 Desert sand 0.40 Grassland / crops 0.20–0.25 Eucalypt forest 0.08–0.12 Open ocean 0.05–0.06 Earth's overall ave

Surface	Typical albedo	Reflectivity
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Eucalypt forest	0.08–0.12	
Open ocean	0.05–0.06	

Albedo lab: melt the ice, feel the warming

03 Interactive · ice–albedo feedback Albedo lab: melt the ice, feel the warming Drag the slider to melt the sea ice off Cape Wirreanda. As bright ice gives way to dark ocean, watch the reflected rays thin out, the absorbed rays multiply, and the surface temperature climb. This is a positive feedback loop in action. 3D · LIVE Sea-ice cover vs absorbed energy 3D view unavailable — use the readouts below. Sea-ice cover 80% Surface albedo 0.62 Energy absorbed 38% Local temp –4.0°C ♦ Analogy — black vs white t-shirt On a hot January day you wear white , not b

The natural greenhouse effect

04 Re-emitted energy The natural greenhouse effect Here's the puzzle: if shortwave sunlight passes straight through the atmosphere, why doesn't all the heat just escape straight back out again? The answer is the change in wavelength when energy is re-emitted. Follow the energy at Cape Wirreanda: Shortwave in: Sunlight passes through the atmosphere (greenhouse gases are transparent to it) and is absorbed by the land and ocean. Surface warms and re-emits the energy — but now as longwave (infrared) radiation, because the surface is far cooler than the Sun.

Greenhouse lab: trap the heat

⚠ EXAM TRAPS

05 Interactive · longwave trap Greenhouse lab: trap the heat Toggle the greenhouse-gas layer on and off. With it off, longwave photons fly straight to space and the surface stays frigid. Switch it on and watch some photons get absorbed and bounced back down — and the temperature readout climb toward a liveable +15 °C. 3D · LIVE Longwave photons escaping vs trapped 3D view unavailable — use the toggle and readout. Greenhouse gases: ON GHG concentration moderate Surface temp +15°C

Ocean circulation: the planet's heat conveyor

06 Moving the energy around Ocean circulation: the planet's heat conveyor Albedo and the greenhouse effect decide how much solar energy Earth keeps. Ocean circulation decides where it goes. Oceans absorb a huge share of incoming sunlight and — because water holds heat far better than air or land — they act as a vast thermal store, then carry that heat thousands of kilometres around the globe. Two systems move the heat Surface currents Driven by prevailing winds. Off Cape Wirreanda, the warm East Australian Current carries tropical heat south down the coast

The three working together at Cape Wirreanda

07 Synthesis The three working together at Cape Wirreanda A single sunbeam arriving at the observatory tells the whole story: ① Albedo decides If it lands on snow it's reflected and lost. If it lands on dark ocean it's absorbed and kept. Albedo sets how much energy stays. ② Greenhouse delays The absorbed energy is re-emitted as longwave, then trapped and recycled by greenhouse gases, raising the surface temperature ~33 °C above frozen. ③ Oceans distribute Currents carry the kept heat around the planet, storing it and spreading it from equator to poles so

Six exam traps

⚠ EXAM TRAPS

08 Don't lose easy marks Six exam traps ⚠ Trap 1 — "Albedo is heat / temperature" Albedo is reflectivity — the proportion of sunlight reflected. It is not a temperature or an amount of heat. Define it as a fraction or percentage from 0 to 1. ⚠ Trap 2 — "Greenhouse gases trap incoming sunlight" They don't. Greenhouse gases are transparent to incoming shortwave. They absorb the outgoing longwave (infrared) the surface re-emits. The wavelength change is the whole mechanism — say it explicitly. ⚠ Trap 3 — Confusing the natural and enhanced greenhouse effect

Command-word decoder

◆ COMMAND WORDS

09 Decode the question Command-word decoder The verb in the question tells you how much to write. Tap each to see what the examiner actually wants. Identify / State Name it, one line. "Snow has a high albedo." No explanation needed — just the fact. Describe Say what happens, in order. Walk through the steps: shortwave in → absorbed → re-emitted as longwave → absorbed by GHGs → re-emitted downward. No need to justify why . Explain Give the cause-and-effect why . Link each step to its consequence: "Because the surface is cooler than the Sun, it re-emits as

Sort it out

◇ ACTIVE RECALL

10 Active recall Sort it out Drag each statement into the process it belongs to. Three mechanisms, eight statements. ALBEDO EFFECT GREENHOUSE EFFECT OCEAN CIRCULATION Fresh snow reflects ~80% of sunlight CO₂ absorbs outgoing infrared radiation The East Australian Current carries warm water south Melting ice exposes darker ocean Back-radiation warms the surface a second time Cold, salty water sinks near Antarctica Keeps Earth ~33°C warmer than it would be A fraction from 0 (dark) to 1 (white) s

P·E·L writing trainer

📝 P·E·L WRITING

11 Build a top-band answer P·E·L writing trainer VCE markers reward structure. The P·E·L method turns a vague paragraph into a mark-grabbing one. Use it on every "explain" question. Practise it now Question: Explain how the albedo effect influences the amount of solar energy Earth retains. (3 marks) P Point — answer the question directly State the relationship up front: albedo is the proportion of sunlight reflected, and it sets how much energy is kept vs lost. E Evidence — give the mechanism + a number/example Contrast a high-albedo and a low-albedo sur



The P·E·L answer structure — make a Point, support with Evidence, then Link to the question.

Weak vs strong: a worked answer

12 See the difference Weak vs strong: a worked answer Question: Explain the natural greenhouse effect and why it is important for life on Earth. (4 marks) **X** Weak response 1 / 4 Greenhouse gases trap the sun's heat in the atmosphere like a greenhouse. This warms up the Earth. Without it the Earth would be cold. Gases like CO₂ cause the greenhouse effect. Vague and mechanism-free. "Trap the sun's heat" misses the shortwave-in / longwave-out distinction — the actual mechanism. No mention of absorption and re-emission, no back-radiation, no figures. "Earth



The P·E·L answer structure — make a Point, support with Evidence, then Link to the question.

Practice exam questions

★ PRACTICE & SELF-MARK

13 Exam-style · 20 marks Practice exam questions Five VCAA-style questions. Try each one first, then reveal the marking guide and a weak vs strong sample. Mark yourself honestly — your score builds in the dock at the bottom right. 1 2 marks Define · Describe Define albedo and explain how the albedo of a surface affects the amount of solar energy that surface absorbs. Reveal answer & mark scheme Marking guide 1 mark — Albedo = the proportion (fraction/percentage, 0–1) of incoming solar (shortwave) radiation that a surface reflects. 1 mark — A higher albed

★ Good vs bad worked answers

Study the contrast — the same question answered two ways. The strong column shows the moves that earn the marks.



The P-E-L answer structure — make a Point, support with Evidence, then Link to the question.

✗ WEAK ANSWER

Greenhouse gases trap the sun's heat in the atmosphere like a greenhouse. This warms up the Earth. Without it the Earth would be cold. Gases like CO₂ cause the greenhouse effect. Vague and mechanism-free. "Trap the sun's heat" misses the shortwave-in / longwave-out distinction — the actual mechanism. No mention of absorption and re-emission, no back-radiation, no figures. "Earth would be cold" isn't specific enough to score. Earns 1 mark for the basic idea that gases retain heat and warm Earth.

✓ STRONG / MODEL ANSWER

Incoming shortwave solar radiation passes through the atmosphere and is absorbed by Earth's surface, which warms and re-emits the energy as longwave (infrared) radiation. Greenhouse gases such as water vapour and CO₂ absorb this outgoing longwave radiation and re-emit it in all directions, including back toward the surface, retaining heat in the lower atmosphere. This keeps Earth's average temperature around +15 °C instead of about -18 °C, making the planet warm enough to support liquid water and life. Four distinct marks: (1) shortwave absorbed by surface; (2) re-emitted as longwave; (3) GHGs absorb and re-emit/back-radiate; (4) quantified importance for life. Each highlighted clause is

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Vague and mechanism-free. "Trap the sun's heat" misses the shortwave-in / longwave-out distinction — the actual mechanism. No mention of absorption and re-emission, no back-radiation, no figures. "Earth would be cold" isn't specific enough to score. Earns 1 mark for the basic idea that gases retain heat and warm Earth.

✓ STRONG / MODEL ANSWER

Four distinct marks: (1) shortwave absorbed by surface; (2) re-emitted as longwave; (3) GHGs absorb and re-emit/back-radiate; (4) quantified importance for life. Each highlighted clause is a scoring point.

✗ WEAK ANSWER

Albedo is how shiny something is. Shiny things don't get as hot. "How shiny" is too loose and never defines albedo as a proportion of reflected solar radiation. No clear link to absorbed energy.

✓ STRONG / MODEL ANSWER

Albedo is the proportion of incoming solar radiation that a surface reflects, on a scale from 0 to 1. A surface with a high albedo (e.g. snow, ~0.8) reflects most sunlight and absorbs little, so it stays cool, whereas a low-albedo surface (e.g. ocean, ~0.06) reflects little and absorbs most of the incoming energy. Definition as a proportion + the absorb/reflect relationship with an example. Full marks.

✗ WEAK ANSWER

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✓ STRONG / MODEL ANSWER

Definition as a proportion + the absorb/reflect relationship with an example. Full marks.

✗ WEAK ANSWER

The sun's rays come in and the greenhouse gases trap them in the atmosphere so the heat can't get out, which warms the Earth up. "Trap the rays" conflates shortwave and longwave and misses that the radiation is re-emitted at a longer wavelength. No back-radiation. Scores the entry-level idea only.

✓ STRONG / MODEL ANSWER

Shortwave solar radiation passes through the atmosphere and is absorbed by the surface, which warms and re-emits the energy as longwave infrared radiation. Greenhouse gases such as CO₂ and water vapour absorb this outgoing longwave radiation and re-emit it in all directions, including back down to the surface, which retains heat in the lower atmosphere. All four steps, with the shortwave→longwave conversion and back-radiation explicit.